

Exercise 2: E-commerce Platform Search Function

1. Understanding Asymptotic Notation

Big O Notation

Big O notation is used to measure the efficiency of an algorithm by describing how its execution time grows as the input size increases.

Search Operation Cases

Best Case

- Linear Search: $O(1)$
- Binary Search: $O(1)$

Average Case

- Linear Search: $O(n)$
- Binary Search: $O(\log n)$

Worst Case

- Linear Search: $O(n)$
- Binary Search: $O(\log n)$

2. Analysis

Time Complexity Comparison

Algorithm	Best Case	Average Case	Worst Case
Linear Search	$O(1)$	$O(n)$	$O(n)$
Binary Search	$O(1)$	$O(\log n)$	$O(\log n)$

Suitable Algorithm for E-commerce Platform

Binary Search is more suitable for an e-commerce platform because it performs searches faster on sorted data with a time complexity of $O(\log n)$, whereas Linear Search requires $O(n)$ time in the average and worst cases.